

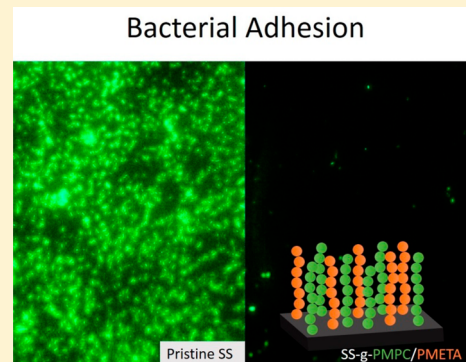
Antifouling and Antimicrobial Coatings from Zwitterionic and Cationic Binary Polymer Brushes Assembled via “Click” Reactions

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Supporting Information

ABSTRACT: Controlled architecture of bifunctional polymers on surfaces is highly challenged because of the stringent reaction conditions or tedious operations required for surface modification. Herein, a simple and effective method was developed to assemble zwitterionic and cationic binary polymer brushes onto polydopamine-anchored stainless steel (SS) surfaces. Zwitterionic poly(2-methacryloyloxyethyl phosphorylcholine) (PMPC) was first graft polymerized from the functionalized SS surface via thiol–ene “click” reaction. Alkynyl-modified cationic poly(2-(methacryloyloxy) ethyl trimethylammonium chloride) (alkynyl-PMETA) was subsequently introduced via azide–alkyne “click” reaction. After the grafting of PMPC/PMETA binary polymer brushes, the resulting functionalized SS surfaces can cooperatively reduce the adhesion of Gram-positive bacteria *Staphylococcus aureus* (*S. aureus*) and Gram-negative bacteria *Pseudomonas* sp., as well as the attachment of microalgae *Amphora coffeaeformis*. In addition, the binary polymer brushes coatings were ascertained to be stable and durable after 30-day exposure to filtered seawater. Thus, surface functionalization with zwitterionic and cationic binary polymer brushes offers an environmentally friendly alternative for biofouling inhibition in the marine and aquatic environments. In addition, surface modification via dual “click” reactions provides another alternation for developing surface coatings with multifunctionalities.



1. INTRODUCTION

Biofouling, also known as the unwanted attachment of protein, bacteria and macro-organisms on materials surfaces, has resulted in economic and ecological consequences in the marine and aquatic environments.^{1–4} Concomitantly, the subsequently formed biofilms and fouling deposits can affect the function of various structures, such as ship hulls, fish nets, submerged pipes and marine platforms.^{2,5} Driven by increasing problems induced by biofouling, the past decade has witnessed the emergence of diverse strategies to inhibit the initial microadhesion of marine organisms.^{6–11} One notable example is the deposition of functional polymer coatings to confer substrates with antiadhesion efficacies. The approach has proven to be effective in both clinical healthcare and marine antifouling applications.^{6,8,12}

Although antimicrobial coatings can efficiently eradicate a variety of microorganisms, the dead organisms settling on coatings will block the antimicrobial functional groups, resulting in an immune response and inflammation.^{13–15} Nonfouling zwitterionic coatings were reported to dramatically resist protein adsorption, bacterial adhesion and biofilm formation, but the pathogenic microbes can still be possibly introduced during implantation operations, resulting in failure of the implanted devices.^{16,17} Thus, it is necessary and desirable to develop surface coatings that have both antimicrobial and nonfouling capabilities. To date, coatings combining antimicrobial and antifouling properties are routinely achievable by the

conventional approaches, including “grafting-to” and “grafting-from” methods.^{6,12,16} In contrast, some recent multilayer coatings have been developed as fouling-resistant marine coatings with effective antimicrobial performance.^{18–20} However, differing from coatings fabricated by the one-step process, deposition of multilayer coatings, e.g., by layer-by-layer deposition, is highly time-consuming,²¹ especially for covalently cross-linked multilayers with a dipping time ranging from 20 min to several hours.

Recently, polymer brushes-based strategies have been developed and applied in various fields.^{22–25} For example, hydrophilic multiwalled carbon nanotubes (MWCNTs) incorporating water-soluble polymer brushes were shown to efficiently improve the mechanical strengths and water uptake of polymer hydrogels.²⁶ As a result, engineering binary polymer brushes as surface coatings has the potential to fulfill the demand for bifunctional surface coatings.

The main challenge of assembling binary polymer brushes on substrates surface is the difficulty to functionalize two different types of polymer chains with high grafting density while maintaining the polymer brush structure. Over the past few years, researchers have contributed to explore effective means

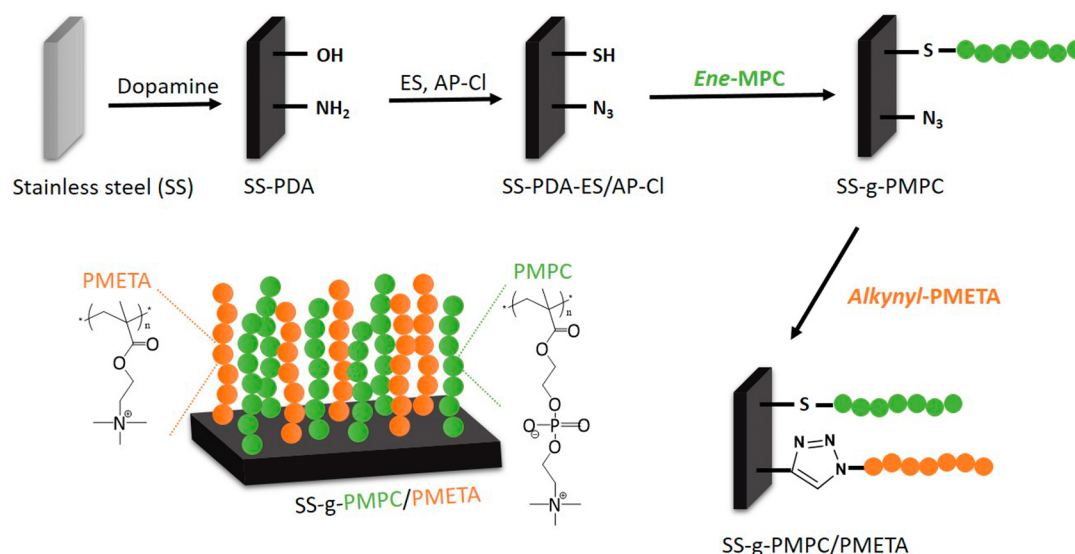
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Scheme 1. Schematic Illustration of Fabricating the Binary Polymer Brushes Coatings on Stainless Steel Surface via “Click” Reaction



that enable the simultaneous functionalization of binary polymer brushes on substrate surface.^{27–29} Employment of two-step surface-initiated control radical polymerization (SI-CRP)³⁰ and functionalization using “click” reactions³¹ have proven to be an effective technique. However, modification using two times of SI-CRP is too complicate and time-consuming. On the other hand, “click” reactions are commonly used for polymer functionalization and macromolecular synthesis because of their attractive properties, such as fast rate, high efficiency and regioselectivity in organic medium at room temperature.^{32–34} Thus, it is desirable to fabricate binary polymer brushes-assembled surfaces by employing two specific “click” reactions, such as thiol–ene and alkyne–azide “click” reactions.

Herein, binary polymer brush coatings with antimicrobial and antifouling efficacies have been prepared. The binary polymer brushes were assembled on SS surfaces via thiol–ene photopolymerization and alkyne–azide “click” reactions consecutively. The SS surface was first deposited with polydopamine (PDA), which serves as the anchor layer (Scheme 1). Reaction of the resulting SS-PDA surface with ethylene sulfide and 3-azidopropyl carbonylimidazole introduces the respective thiol groups and azide groups (SS-PDA-ES/AP-CI). Zwitterionic 2-methacryloyloxyethyl phosphorylcholine (MPC) was subsequently grafted polymerized from the thiol-functionalized SS surface via thiol–ene photopolymerization (the SS-g-PMPC surface). Antimicrobial cationic polymer brushes of poly(2-(methacryloyloxy)ethyl trimethylammonium chloride) (alkynyl-PMETA) with alkynyl functionalities were synthesized via ATRP and then clicked onto the azide functionality of the SS surface to complete the bifunctional SS-g-PMPC/PMETA brush coatings. The antifouling performance of the resulting binary polymer brushes-functionalized SS surfaces was assayed against bacterial adhesion (both *S. aureus* and *Pseudomonas* sp. bacteria), as well as microalgal attachment (*Amphora coffeaeformis*). In addition, the cytotoxicity of the resulting binary polymer brushes coatings to 3T3 fibroblasts was also evaluated.

2. EXPERIMENTAL SECTION

2.1. Materials. Stainless steel foils (SS, AISI-type 304, 0.05 mm thick) were purchased from Goodfellow Ltd. of Cambridge, U.K. Dopamine hydrochloride (98%), Tris-(hydroxymethyl) aminomethane hydrochloride (Tris–HCl), ethylene sulfide (ES, 98%), 2-methacryloyloxyethyl phosphorylcholine (MPC, 97%), 1,1'-carbonyldiimidazole (CDI), copper(I) bromide (CuBr, 99%), 2-(methacryloyloxy)ethyl trimethylammonium chloride solution (META, 80 wt % in H₂O), propargyl bromide and 2,2'-bipyridyl were purchased from Sigma-Aldrich Co., St. Louis, MO. The LIVE/DEAD BacLight Bacterial Viability Kit L131152 was purchased from Molecular Probes Inc., Eugene, Oregon, USA. All other reagents and solvents were purchased from either Sigma-Aldrich or Merck Chem. Co. without further purification. 3-Azidopropyl carbonylimidazole (AP-CI) and alkynyl-PMETA were synthesized following methods published previously.³⁵ *Staphylococcus aureus* (*S. aureus*, ATCC 12228, Gram-positive) were obtained from American Type Culture Collection, Manassas, VA. *Pseudomonas* sp. (NCIMB 2021, Gram-negative) were obtained from the National Collection of Industrial Marine Bacteria, Sussex, U.K.

2.2. Immobilization of Polydopamine Anchor on SS Surfaces. Prior to the immobilization of dopamine, 2 × 2 cm newly cut SS foils were washed in Decon 90 in an ultrasonic bath for 5 min, and then activated in a piranha solution (50 mL of H₂O₂ to 150 mL of H₂SO₄) at room temperature (25 °C) for 30 min, to obtain the pristine SS surfaces. Next, the pristine SS surfaces were immersed in 30 mL of Tris–HCl (10 mM, pH 8.5), containing 60 mg of dopamine hydrochloride. The reaction mixture was left on a shaking device for 5 h at 37 °C. After the reaction, the polydopamine anchored SS substrates (SS-PDA) were rinsed with DI water thoroughly and dried with compressed air.

2.3. Preparation of “Clickable” SS Surfaces (SS-PDA-ES/AP-CI). The “clickable” SS surface with thiol groups (SS-PDA-ES) was first prepared. Briefly, SS-PDA substrates were introduced in a reaction flask with 5 mL of ethanol. After degassing with argon for 30 min, ES (50 μL, 0.42 mmol) was dropwise added, the reaction mixture was kept at room

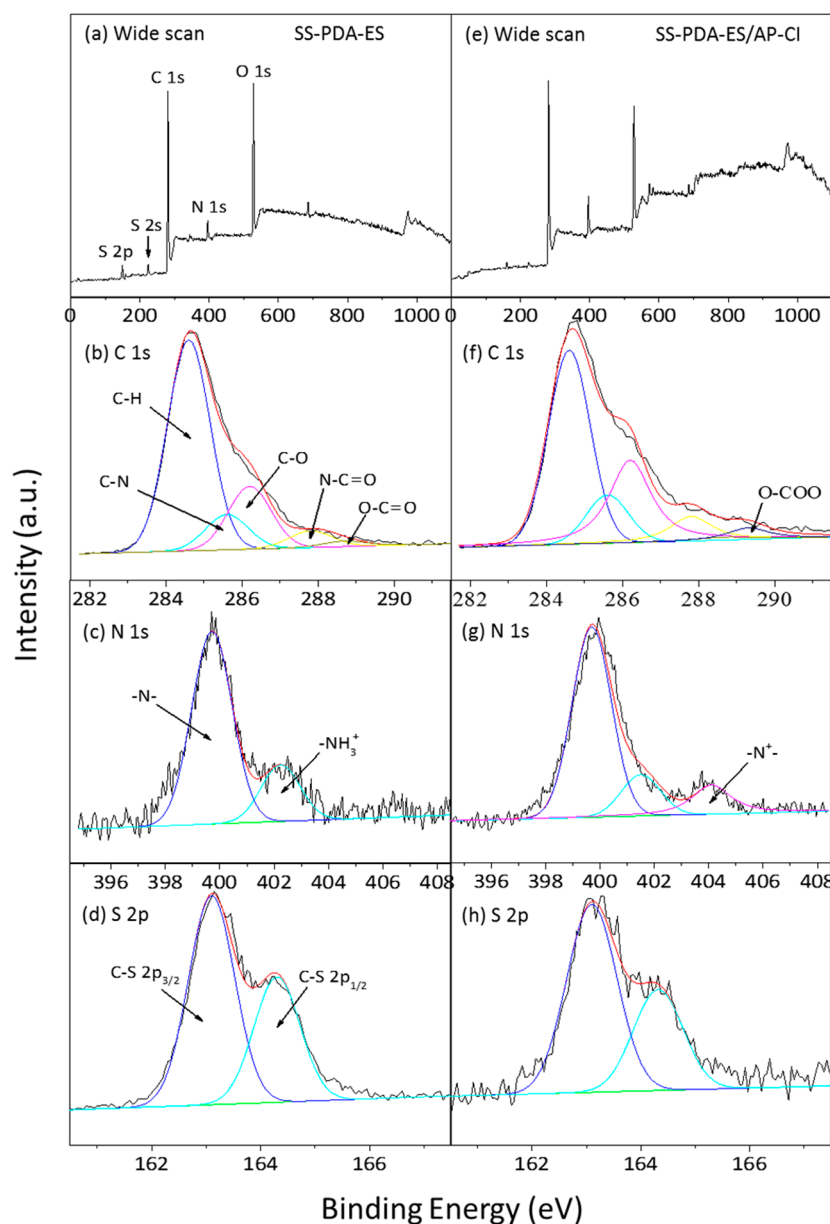


Figure 1. XPS wide-scan, C 1s, N 1s and S 2p core-level spectra of the (a–d) SS-PDA-ES and (e–h) SS-PDA-ES/AP-CI surfaces.

temperature for 1 h. Following the completion of the reaction, the SS-PDA-ES surface were thrice washed with ethanol and DI water, and then dried with compressed air.

Subsequently, to prepare the “clickable” SS surface (SS-PDA-ES/AP-CI) with both thiol and azide groups, the prepared SS-PDA-ES substrates were immersed in a mixture of toluene (15 mL) and the synthesized AP-CI (600 μ L) in a reaction flask. The reaction was kept in an oil bath at 60 $^{\circ}$ C for 24 h. The as-obtained SS-PDA-ES/AP-CI substrates were washed with toluene and ethanol, and then dried with compressed air.

2.4. PMPC Functionalization via Thiol–Ene “Click” Photopolymerization. Surface-initiated thiol–ene “click” photopolymerization of methacryloyloxyethyl phosphorylcholine (MPC) on the SS-PDA-ES/AP-CI surface was carried out according to the methods reported previously.^{36,37} 2 mL of DI water, 120 mg of MPC and the SS-PDA-ES/AP-CI substrates were added in a Pyrex glass tube, and the mixture solution was continuously degassed with argon flow for 40 min (30 min in solution and 10 min above solution). Then, the glass tubes

were sealed and placed in Riko rotary photochemical reactor (RH400-10W) with UV light for 1 h. The zwitterionic PMPC-functionalized SS substrates (SS-g-PMPC) were obtained after washing with DI water and drying with compressed air.

2.5. PMETA Functionalization via Azide–Alkyne “Click” Reaction. To prepare the binary polymer brushes-coated SS surfaces (SS-g-PMPC/PMETA), the SS-g-PMPC substrates were added in 5 mL of DI water, containing 1.5 mg of copper(II) sulfate, 3.6 mg of sodium ascorbate and 60 mg of the synthesized alkynyl-PMETA. The reaction mixture was sealed and placed into the shaker at 30 $^{\circ}$ C for 6 h. After the completion of the reaction, the solution turned from yellow to colorless. Then, the obtained SS-g-PMPC/PMETA surfaces were thoroughly washed with DI water and dried with compressed air.

2.6. Antifouling Assays of the Binary Polymer Brushes-Functionalized Coatings. **2.6.1. Bacteria Adhesion Assays.** Two bacterial strains, *S. aureus* and *Pseudomonas* sp., were employed to evaluate the antiadhesion performance of

the pristine and modified SS surfaces. Culture and incubation of these two bacteria were proceeded according to previously published literature.¹⁶ The concentration of the testing bacteria is 10^7 cells/mL. Qualitative and quantitative analysis of adhered bacteria on the pristine and functionalized SS surface were carried out by fluorescence microscopy images and spread plate method, respectively, as reported in previous literatures.^{16,38}

2.6.2. Microalgal Attachment Assays. *Amphora coffeaeformis* (UTEX B2080) was cultured and employed to evaluate the antiadhesion performance of the binary polymer brushes coating against microalgae according to procedures described previously.²⁰ Quantification of attached *Amphora* cells on the pristine and polymer brushes-coated SS surfaces was performed by determining the autofluorescence intensity of detached cells, as reported in previous literature.¹⁶ The results were expressed as adhered *Amphora* cell percentages relative to that of the pristine SS surface.

2.7. Cytotoxicity Test of the Binary Polymer Brushes-Functionalized Coatings. Cytotoxicity of the pristine, SS-PDA-ES/AP-CI, SS-g-PMPC and SS-g-PMPC/PMETA substrates were evaluated by immersing each sample substrate (1×1 cm in size) into 2 mL of DMEM medium at 4°C for 3 and 10 days. After that, the medium pretreated with each sample substrate was collected and assayed with 3T3 fibroblasts culture (ATCC, Manassas, USA) via MTT assays.³⁸ In addition, the original culture medium, without exposure to sample substrates, was employed as the control.

2.8. Stability and Durability Assays. The stability and durability of the bifunctional coatings were assayed by subjecting the functionalized SS surfaces to fresh seawater ($0.2\ \mu\text{m}$ filter) for 30 days, termed as the aged SS-g-PMPC/PMETA surface. The elemental composition of these aged surfaces was determined by XPS. The remaining antimicrobial and antifouling performance were evaluated by bacterial adhesion and microalgal attachment using fluorescence microscopy images.

3. RESULTS AND DISCUSSION

3.1. Characterization of AP-CI and Alkynyl-PMETA. 3-Azidopropyl carbonylimidazole (AP-CI) was synthesized to introduce azide groups on the polydopamine-deposited SS (SS-PDA) surface for the subsequent alkynyl–azide “click” reaction. ^1H Nuclear magnetic resonance (^1H NMR) spectroscopy was used to confirm the successful synthesis of AP-CI. As shown in Figure S1 (Supporting Information), ^1H NMR (CDCl_3) spectroscopy results: δ (ppm) 8.12 (1H, s, $\text{N}-\text{CH}=\text{N}$), 7.40 (1H, s, $\text{N}-\text{CH}=\text{C}$), 7.06 (1H, s, $\text{C}=\text{CH}-\text{N}=\text{}$), 4.50 (2H, t, CH_2-O), 3.47 (2H, t, N_3-CH_2), 2.05 (2H, app. quin, $\text{C}-\text{CH}_2-\text{C}$), which is consistent with that reported previously.³⁹ The average molecular weight of the as-synthesized alkynyl-PMETA was determined by GPC and found to be about 12000 g/mol, indicating the successful polymerization of alkynyl-PMETA via ATRP.

3.2. Functionalization of SS-PDA with ES and AP-CI for “Clickable” Surface. XPS was used to evaluate the step-by-step functionalization on SS surfaces by determining the respective chemical compositions. Figure 1 shows the XPS wide-scan, C 1s, N 1s and S 2p core-level spectra of the SS-PDA-ES and SS-PDA-ES/AP-CI surfaces, respectively. Differing from the XPS spectrum of the SS-PDA surface (Figure S2a), after functionalization of the ES molecules that react readily with the amine groups in polydopamine, two apparent XPS signals appear in that of the SS-PDA-ES surface (Figure

1a), which are attributable to S 2p (165 eV) and S 2s (228 eV) signals of the introduced thiol groups.^{40,41} In addition, in the S 2p XPS core-level spectrum of the SS-PDA-ES surface, a strong spin–orbit splitting doublet can be observed at binding energies (BEs) of 163.1 and 164.3 eV (Figure 1d), indicative of the S $2p_{3/2}$ and S $2p_{1/2}$ signals respectively.^{40,42} These BEs on the SS-PDA-ES surface are attributed to the successful modification of thiol groups on the SS-PDA surface.

Subsequently, AP-CI was introduced to the SS-PDA-ES surface through the reactions with the hydroxyl and amine groups. In the C 1s core-level spectrum of the SS-PDA-ES/AP-CI surface (Figure 1f), an additional XPS peak component with a BE of 290.5 eV is observed, which is consistent with the presence of carbonate ester linkage ($\text{O}-\text{COO}$)^{40,43} formed from the reaction of AP-CI with the SS-PDA-ES surface. Additionally, a new peak component at the BE of 404.1 eV appears in N 1s core-level spectrum of the SS-PDA-ES/AP-CI surface (Figure 1g), attributable to the positively charged nitrogen (N^+) in the azide groups.⁴⁰ The new peak component of $\text{O}-\text{COO}$ species in Figure 1f, as well as the newly appeared N^+ species in Figure 1g, confirms the successful preparation of the SS-PDA-ES/AP-CI surface by the functionalization of azide groups. Besides, the persistence of thiol groups is evidenced by the peak components at the BEs of 165 and 228 eV in Figure 1e. Thus, a “clickable” surface with both thiol groups and azide groups has been prepared.

3.3. Binary Polymer Brushes Coatings Prepared via “Click” Reactions. Zwitterionic 2-methacryloyloxyethyl phosphorylcholine (MPC) was photopolymerized and grafted from the SS-PDA-ES/AP-CI surface via thiol–ene “click” reaction (Figure 2a–c). In Figure 2a, the P 2s and P 2p core-level signals, with respective BE of 188 and 131 eV^{40,44} are

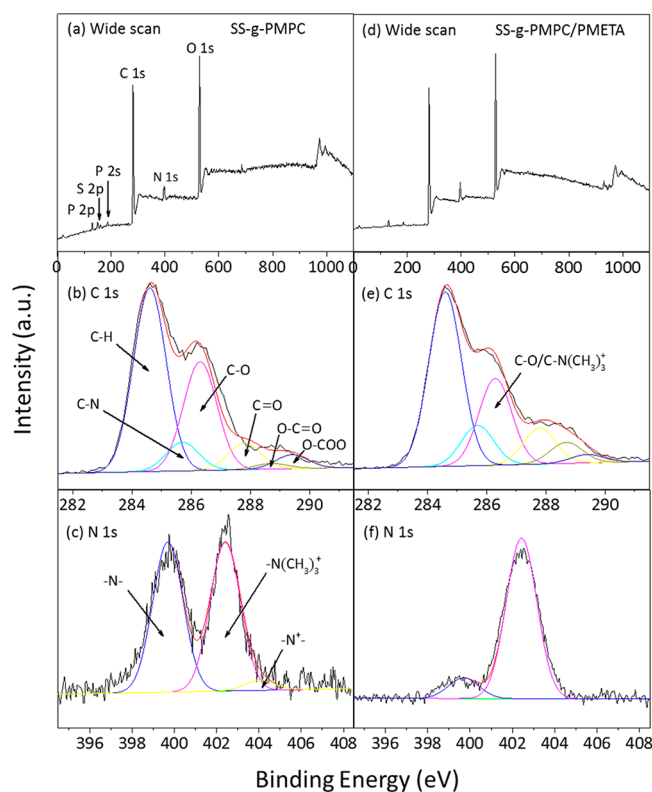


Figure 2. XPS wide-scan, C 1s and N 1s core-level spectra of the (a–c) SS-g-PMPC and (d–f) SS-g-PMPC/PMETA surfaces.

discernible in the XPS wide-scan spectrum of the SS-g-PMPC surface after functionalization of PMPC polymer brushes. In addition, the N 1s XPS core-level spectrum of the SS-g-PMPC surface contains two peak components with BEs at 399.7 and 402.4 eV (Figure 2c), which can be attributed to the amine ((C)–N) and quaternary ammonium nitrogen ($-\text{N}(\text{CH}_3)_3^+$) species respectively,^{40,45} suggesting the successful coupling of PMPC to the SS-PDA-ES/AP-CI surface via thiol–ene photopolymerization. Also, the XPS signal with BE at 404.1 eV in the N 1s core-level spectrum of the SS-g-PMPC surface (Figure 2c) is assigned to N^+ species in residual azide groups.^{40,46} Thereafter, PMETA brush was “clicked” onto the SS-g-PMPC surface via azide–alkyne “click” reaction (Figure 2d–f). In the N 1s core-level spectrum of the SS-g-PMPC/PMETA surface (Figure 2f), the intensity of $-\text{N}(\text{CH}_3)_3^+$ peak component increases while that of the amine ((C)–N) species decreases significantly. Moreover, the N^+ peak component at 404.1 eV is longer discernible, suggesting that PMETA brushes have been successfully grafted on the SS-g-PMPC surface via azide–alkyne “click” reaction.^{40,46,47} These results indicate the grafting of PMPC/PMETA binary polymer brushes, and thus the successful preparation of the bifunctional polymer coatings, on the SS surfaces. From the XPS-derived surface chemical composition (P 2p and N 1s peak component area ratio) of the SS-g-PMPC/PMETA surface (Figure 2d), and taking into account of the elemental N and P stoichiometries in the MPC and META repeat units, the ratio of PMPC over PMETA brushes on the SS surface was determined to be about 1:2.

To evaluate the stability and durability of binary polymer brush-functionalized surfaces, the SS-g-PMPC/PMETA surface was exposed to filtered fresh seawater for 30 days, and then the XPS spectra of the aged surface were determined. No appreciable changes are observed in the XPS wide-scan and C 1s core-level spectra of the aged SS-g-PMPC/PMETA surface (Figure S2b), in comparison to that of the SS-g-PMPC/PMETA surface (Figure 2), demonstrating that the binary polymer brushes-modified coatings fabricated by thiol–ene photopolymerization and azide–alkyne “click” reactions are stable and durable.

3.4. Surface Properties of the Polymer Brushes-Functionalized Surfaces. Surface properties, such as morphology, thickness, wettability and surface charge are known to dramatically affect the interactions between micro-organisms and substrate surfaces.^{48,49} To assess the surface wettability, the static water contact angles (WCAs) of the pristine and brush-coated SS surfaces were determined (Figure 3). The WCA increases slightly from 66° to 69°, after coupling of PDA, ES and AP-CI. However, with the introduction of PMPC on the SS-PDA-ES/AP-CI surface via thiol–ene photopolymerization, the hydrophilicity of the SS-g-PMPC surface is dramatically enhanced, with a decreased WCA decreases from 69° to 36°, consistent with the presence of a hydration layer on the SS-g-PMPC surface.⁵⁰ Subsequent coupling of PMETA gives an even more hydrophilic surface with a WCA as small as 31°, indicative of the presence of a stronger surface hydration on the binary polymer brush-functionalized surface. Moreover, the aged SS-g-PMPC/PMETA surface still retains a small WCA of 35°, confirming the stability and durability of the grafted coatings after 30 days of exposure to seawater. Subsequently, the surface morphologies of functionalized glass slide (GS) surfaces were also investigated by AFM, as shown in Figure 4. In comparison to the GS-PDA-ES/AP-CI surface, the root-mean-square rough-

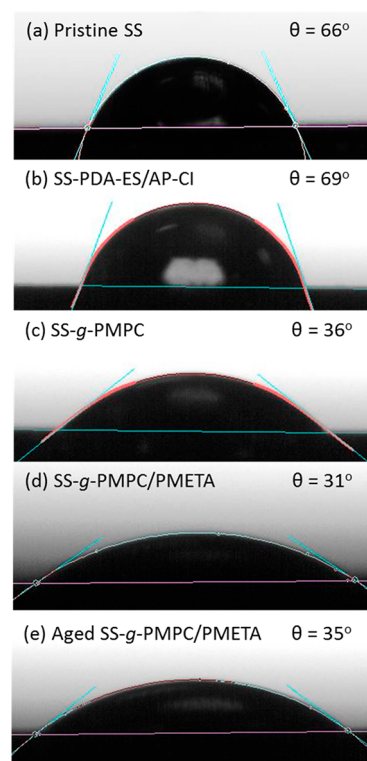


Figure 3. Images of water droplets on the (a) pristine SS, (b) SS-PDA-ES/AP-CI, (c) SS-g-PMPC, (d) SS-g-PMPC/PMETA and (e) aged SS-g-PMPC/PMETA surfaces.

ness (R_q) of the GS-g-PMPC surface over an area of $1 \times 1 \mu\text{m}$ increases from 0.9 to 1.6 nm. The increase in surface roughness results from the grafted PMPC polymer brushes via thiol–ene photopolymerization. On the other hand, after the subsequent functionalization of PMETA polymer brush via azide–alkynyl “click” reaction, the surface roughness decreases to 1.24 nm, indicating the formation of a smoother surface arising from denser coatings by the binary brushes.

In addition to surface wettability and morphology, the coating thickness and surface charge of functionalized GS/SS surfaces were also evaluated to further confirm the fabrication of binary polymer brushes coating (Figure S3, Supporting Information). The coating thickness of the “clickable” surface (GS-PDA-ES/AP-CI) is ~ 40 nm. After functionalization of the PMPC brushes, the thickness increases to about 52 nm, indicating the successful polymerization of PMPC polymer brushes. The subsequent grafting of PMETA polymer brushes is revealed by the further increased coating thickness (from 52 to 59 nm). In addition, ζ -potential of the modified SS surfaces was also measured to confirm the sequential grafting of PMPC and PMETA brushes. The SS-PDA-ES/AP-CI surface is negatively charged at pH 7.4, with a ζ -potential of -28 mV. Functionalization of surface with zwitterionic PMPC brushes markedly alters the charge of SS-g-PMPC surface to a neutral state (-2 mV at pH 7.4). After subsequent grafting of the cationic PMETA brushes, the SS-g-PMPC/PMETA surface has become positively charged (22 mV at pH 7.4), which is in agreement with that reported for the PMETA brushes-grafted surface.⁵¹ The sequential increases in coating thickness and ζ -potential clearly indicate the successful modification of the surface with PMPC/PMETA binary polymer brushes via “click” reactions.

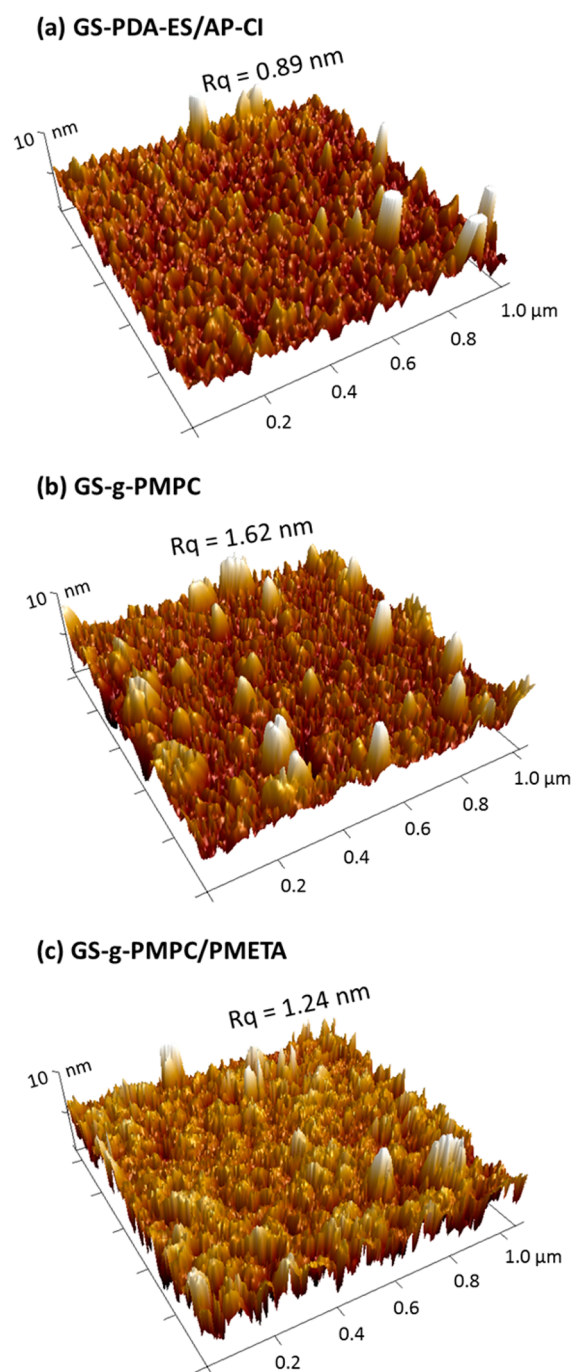


Figure 4. Surface morphologies of the (a) GS-PDA-ES/AP-CI, (b) GS-g-PMPC and (c) GS-g-PMPC/PMETA surfaces determined by AFM. R_q = root-mean-square roughness.

3.5. Bacterial Adhesion Assays. Bacterial adhesion to the artificial surfaces is considered as the predetermining step during biofouling and the biofilm formation.^{52,53} In this study, *S. aureus* and *Pseudomonas* sp. were used to evaluate the antimicrobial and antifouling efficacies of polymer brushes-functionalized SS surfaces. LIVE/DEAD two-color fluorescence-based method, in which the live and dead cells can be distinguished as the live cells appear green and the dead cells appear red under the fluorescence microscopy, was used to evaluate the adhered bacteria. After incubation in the *S. aureus* bacterial suspension (10^7 cells/mL), Figure 5 reveals the fluorescence images of the pristine and polymer brushes-coated

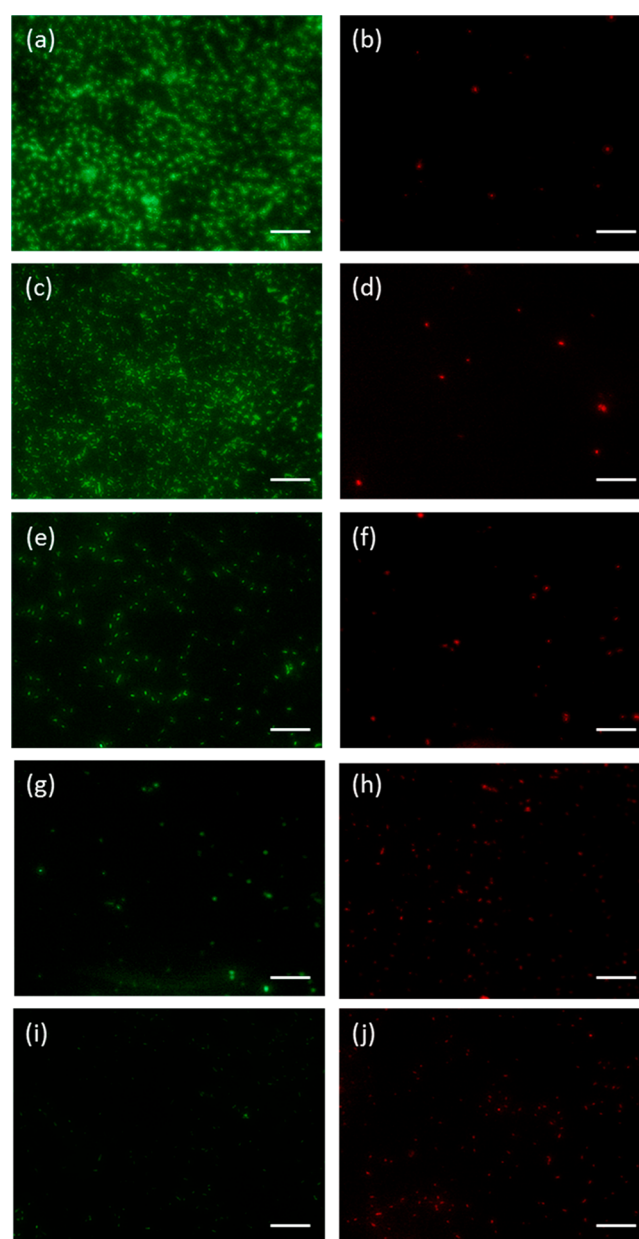


Figure 5. Fluorescence microscopy images of the (a, b) pristine SS, (c, d) SS-PDA-ES/AP-CI, (e, f) SS-g-PMPC, (g, h) SS-g-PMPC/PMETA and (i, j) aged SS-g-PMPC/PMETA surfaces after exposure to *S. aureus* (10^7 cells/mL) for 4 h. Scale bar: 50 μ m.

SS surfaces. A large amount of green spots (viable cells) can be observed in Figure 5a, with only a few dead cells (Figure 5b), indicating the negligible antiadhesion efficiency and low toxicity of the unmodified SS surface. For the SS-PDA-ES/AP-CI surface, the number of adhered bacteria is slightly smaller than that of the pristine SS surface (Figure 5c,d). Notably, after functionalization of the zwitterionic PMPC polymer brushes, the amount of adhered viable cells observed on the SS-g-PMPC surface (Figure 5e,f) is significantly reduced, indicating that the grafted zwitterionic polymer brushes play an important role in resisting bacterial adhesion. After the subsequent functionalization of PMETA polymer brushes, the number of viable bacteria adhered on the SS-g-PMPC/PMETA surface is further reduced (Figure 5g,h). The bacterial resistance of the SS-g-PMPC and SS-g-PMPC/PMETA surfaces can be ascribed to the strong

surface hydration layer induced by the zwitterionic PMPC polymer brushes, as well as the smooth surface morphology. However, there is an obvious increase in dead bacteria (red spots) on the SS-g-PMPC/PMETA surface (Figure 5h). The phenomenon can be attributable to the electrostatically induced lethal effect of the quaternary ammonium cations in PMETA brushes. Moreover, as shown in Figure 5i,j, only a slight increase in the number of attached bacteria is observed on the aged SS-g-PMPC/PMETA surface, in comparison to the surfaces before aging, suggesting the presence of a stable and durable polymer coating.

To quantitatively assess the antiadhesion performance of the modified SS surfaces, the adhered bacteria was counted using the spread plate method.⁴⁶ Figure 6 presents the fractions of

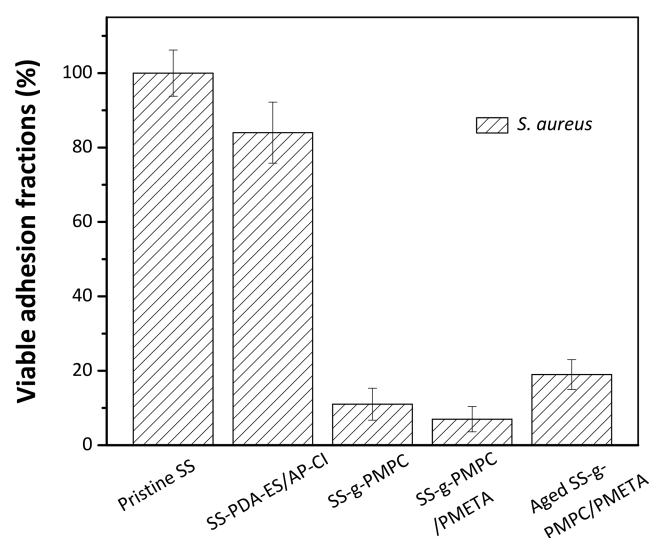


Figure 6. Percentage of the adhered *S. aureus* cells on the pristine, functionalized and aged SS surfaces, after exposure to the bacterial suspension in artificial seawater (10^7 cells/mL) for 4 h. Each error bar represents the standard deviation calculated from three replicates.

adhered *S. aureus* on the pristine and modified SS surfaces after a 4 h of exposure at 37 °C. For the SS-PDA-ES/AP-CI surface, as no antifouling agents have been coated on, the number of viable cells remains high at 84%. The functionalization of PMPC polymer brushes significantly reduces the viable adherent fractions to 11% of the control, and the SS-g-PMPC/PMETA surface shows the lowest cell viability at 7%. The stability and durability of the binary polymer brush coatings are further ascertained as the aged SS-g-PMPC/PMETA surface reduces the adhered viable cells to 19%. These results are in accordance with the fluorescence microscopy images (Figure 5), confirming the antifouling property of the binary polymer brushes coatings.

Pseudomonas sp. is also commonly used to evaluate the antiadhesion performance due to their widespread occurrence in water and marine environments.¹⁶ As shown in Figure S4, similar conclusion of the bacterial adhesion can be obtained by analyzing the fluorescence images of adhered Gram-positive bacteria *Pseudomonas* sp. on the pristine and polymer brushes-coated surfaces, validating the antimicrobial efficiency of the cationic PMETA brushes and the antifouling property of the zwitterionic PMPC brushes. In addition to significantly reduced bacterial adhesion on the SS-g-PMPC/PMETA surfaces, the antifouling and antimicrobial efficacies of the binary polymer

brush coatings are also comparable to other fouling-resistant coatings, including PEG-based coatings,^{54,55} block polymer coatings^{56,57} and multilayer coatings^{18,20} assembled via layer-by-layer deposition.

3.6. Microalgal Attachment Assays. The antiadhesion properties of pristine SS and modified SS substrates were also investigated by *Amphora coffeaeformis* raphid diatom, which is a wide-distributed microfouler in marine environment.^{58,59}

Figure 7 shows the fluorescence images of pristine SS and

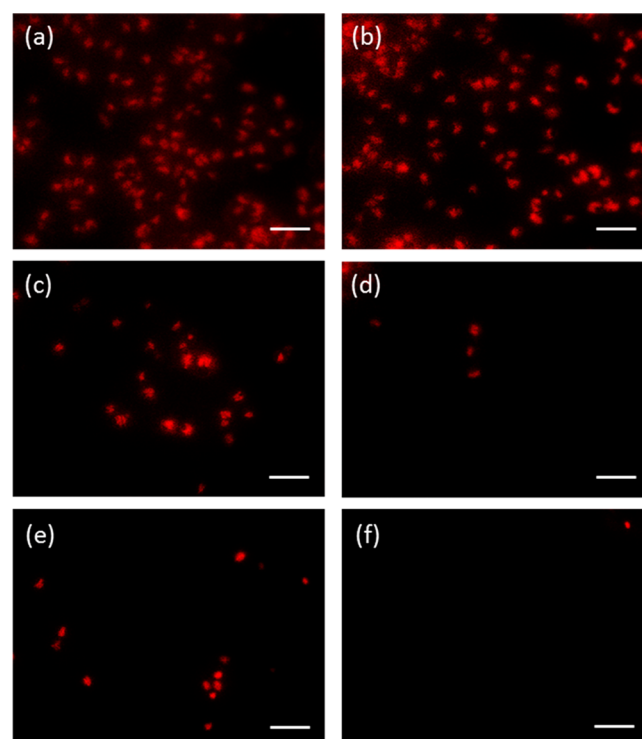


Figure 7. Fluorescence microscopy images of the (a) pristine SS, (b) SS-PDA-ES/AP-CI, (c) SS-g-PMPC, (d) SS-g-PMPC/PMETA and (e) aged SS-g-PMPC/PMETA surfaces after immersion in the *Amphora coffeaeformis* solution at 25 °C for 24 h and (f) the detachment from the pristine SS after 10 min of ultrasonic treatment. Scale bar: 50 μ m.

modified SS surfaces after exposure to *Amphora* suspension for 24 h. A large number of adhered *Amphora* cells is captured on the pristine SS surface (Figure 7a), similar to that of SS-PDA-ES/AP-CI (Figure 7b). In contrast, the number of the adhered *Amphora* cells is significantly reduced on the SS-g-PMPC surface (Figure 7c). The adhered *Amphora* cells are further reduced on the SS-g-PMPC/PMETA surface (Figure 7d), indicating an effective resistance to microfouler *Amphora*. In addition, there is only a small number of *Amphora* adhered to the aged SS-g-PMPC/PMETA surface (Figure 7e), as compared to the pristine SS surface (Figure 7a). The observed *Amphora* adhesion on various SS surfaces are in good agreement with that of bacterial adhesion.

Besides the visual comparisons of *Amphora* adhesion between the different modified surfaces, a more quantitative assay was also performed.⁶⁰ *Amphora* cells were almost detached from the fouled surface, as only one red fluorescence spot of *Amphora* cell remained after 10 min of ultrasonication (Figure 7f). Figure 8 shows the quantitative essay of pristine and modified SS surfaces. The pristine SS surface is chosen as

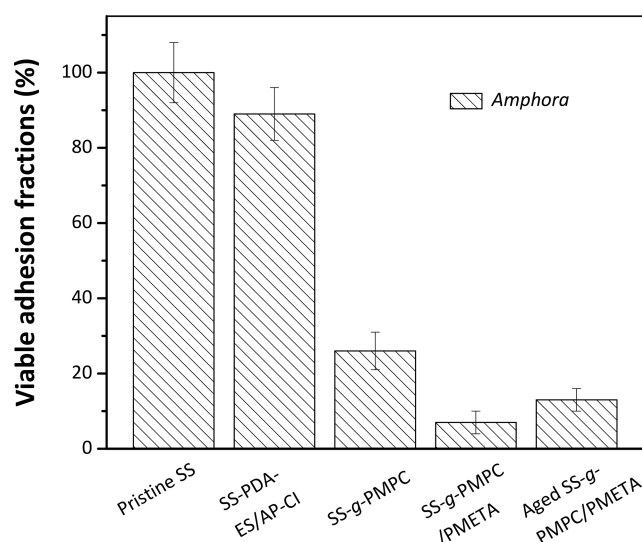


Figure 8. Percentage of settled *Amphora* on the pristine, functionalized and aged SS surfaces after immersion in the 10^5 cell/mL algal suspension for 24 h. Each error bar denotes the standard deviation from three replicates.

control at 100% because it is highly susceptible to the *Amphora* cells (Figure 7a). For the SS-PDA-ES/AP-CI, the amount of attached *Amphora* cells remains high at 89%. The amount of attached *Amphora* cells is significantly reduced on the SS-g-PMPC, to 26%. Functionalization of alkynyl-PMETA onto the SS surfaces further reduced the *Amphora* adhesion to only 7%. Also, the aged SS-g-PMPC/PMETA surface retains the antifouling performance with 13% of attached *Amphora* cells. These results are consistent with the observed fluorescence images (Figure 7).

3.7. Cytotoxicity of the Polymer Brushes Functionalized Surfaces. To evaluate the cytotoxic effect of binary polymer brushes-coated surface, MTT assays with 3T3 fibroblasts were carried out. In Figure 9, after functionalization

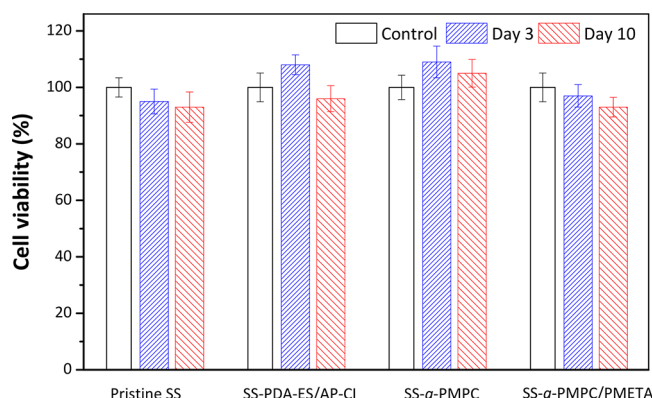


Figure 9. Effect of the pristine SS, SS-PDA-ES/AP-CI, SS-g-PMPC and SS-g-PMPC/PMETA surfaces exposed media (3 and 10 days) on growth of 3T3 fibroblasts measured by MTT assay.

of the “clickable” groups and PMPC brushes, the cell viabilities are higher than that of the pristine SS surface, suggesting of reduced cytotoxicity. The cell viability however decreased slightly after grafting of the PMETA polymer brushes. This phenomenon is ascribed to the quaternary ammonium cations in PMETA, which play a negative role on the growth of 3T3

fibroblasts. Nevertheless, the MTT assay results indicate that the overall biocompatibility of the as-prepared surfaces with binary brush coatings remains high, with cell viabilities of higher than 93% relative to that of the control medium.

4. CONCLUSION

Robust binary polymer brushes coatings have been prepared via thiol–ene and azide–alkyne “click” reactions. The zwitterionic and cationic binary polymer brushes are engineered to impart cooperatively both antifouling and antimicrobial properties. Zwitterionic PMPC brushes were grafted from the SS-PDA surfaces via thiol–ene “click” photopolymerization of MPC, and cationic PMETA polymer brushes were subsequently clicked onto the SS-PDA surfaces via azide–alkyne “click” reaction. Compared to the previously reported methods, this approach is simple, straightforward and effective, providing an alternative to the design of bifunctional polymer coatings. The resulting binary polymer brushes-grafted SS surfaces exhibit good resistance to the bacterial adhesion of *S. aureus* and *Pseudomonas* sp., as well as the microalgal attachment of *Amphora coffeaeformis*. The overall antifouling and antimicrobial efficacies are improved with the employment of binary polymer brushes, instead of only employing the zwitterionic PMPC polymer brushes. In addition, the coatings are proven to be stable, durable and biocompatible. Therefore, the as-prepared binary polymer brushes coatings are potential alternative to combat biofouling in marine environments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.7b03132.

Characterization sections, ^1H NMR spectrum of AP-CI in CDCl_3 ; XPS wide-scan and C 1s spectra of the SS-PDA and aged SS-g-PMPC/PMETA surfaces; ζ -potential and coating thickness of functionalized SS surfaces; fluorescence microscopy images of the pristine, functionalized and aged SS surfaces after exposure to 10^7 cells/mL *Pseudomonas* sp. bacterial suspension (PDF)

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Notes

The authors declare no competing financial interest.

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